

YIXUAN DENG

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Research Interest: Multimodal Representation Learning, Audio-Visual Computing, Generative Modeling, Visual Computing

EDUCATION

Wuhan Textile University

Sep. 2023 – Jun. 2027

Bachelor's Degree (expected), Data Science and Big Data Technology

GPA: 87.31/100 (Top 5%)

PUBLICATIONS & MANUSCRIPTS

Chengkun Rao, **Yixuan Deng**, Ouxiang Li, Ziwei Luo, Yangjun Ou, Zhaojing Wang, Min Li, and Xiaoyun Yan. "Depth-Assisted Skeleton Detection in Natural Images." *ECCV 2026 Conference Submission, under review*. Student paper; Subject area: Low-level vision; Contribution type: Algorithms/General.

Qin Zheng, Yuhang Guan, **Yixuan Deng**, Binju Huang, and Hongyan Wan. "A Statistical Modeling-Based Study of Optimal Memory Curves under Adolescent-Inspired Learning Dynamics." *Submitted to QRS Workshop, under review*.

RESEARCH EXPERIENCE

Depth-Assisted Skeleton Detection in Natural Images

Oct. 2025 – Mar. 2026

Co-author | *ECCV 2026 Conference Submission, under review*

- Co-authored DASkel, the first depth-assisted framework for skeleton detection in natural images, targeting the degradation of RGB-only methods under complex content.
- Studied how depth images provide color- and texture-insensitive structural cues, including regional contours and spatial relationships between regions.
- Worked with a dual-stream RGB-D architecture using separate DINOv3 backbones, MoLA fine-tuning, and a multi-orientation context module for feature selection and integration.
- Contributed to experiments across mainstream skeleton detection datasets, where depth-assisted modelling demonstrated clear performance gains over prior methods.

NLMEM-Transformer-MoE Model for Student Memory Retention Prediction & Personalized Review

Recommendations

Apr. 2025 – Jun. 2025

- Applied Nonlinear Mixed Effects Models to analyze individual learning differences and memory retention patterns.
- Constructed a coefficient-gated Mixture of Experts Transformer architecture for personalized review recommendation.
- Implemented multi-head self-attention and temporal encoding layers to model sparse timestamp data and sequential learning behaviours.
- Completed data preprocessing workflows and supported model architecture design.
- This project resulted in a manuscript submitted to QRS on statistical and deep-learning-based memory-curve modeling.

Stable Diffusion-Based Garment Editing and Generation System

Mar. 2024 – Jun. 2024

- Fine-tuned Stable Diffusion models for clothing generation and editing tasks under limited training data conditions.
- Introduced LoRA to reduce training cost and integrated DragDiffusion to support interactive image manipulation.
- Refactored frontend interfaces and developed a web-based UI with CSS and JavaScript, connecting model functionality with user-facing workflows.

ENGINEERING PROJECTS

University "Challenge Cup" Registration Management System.

Jan. 2026 – Mar. 2026

Backend Developer

- Developed the backend of a university-wide competition registration and review platform adopted across 13 school units and used in multiple rounds with over 1,000 cumulative user interactions.
- Implemented core services for user registration, team formation, material submission, and multi-role review workflows using Spring Boot, FastAPI, MySQL, and Redis.
- Supported production deployment and ongoing operation by designing database schemas and indexes, handling API exceptions, and improving data consistency and system reliability in live use.

Smart Wearable Measurement Device Data Service System

Jul. 2025 – Sep. 2025

Backend Developer

- Built device data service APIs with FastAPI and developed backend modules for measurement-data processing.

- Implemented OpenCV-based edge detection, line detection, feature extraction, and similarity analysis for automated measurement workflows.
- Improved scalability through modular decomposition and interface decoupling.

INTERNSHIP EXPERIENCE

Jiangxia District Special Education School

Mar. 2025 – Jun. 2025

Computer Teacher

- Delivered programming and computer-usage instruction adapted to students across different grade levels.
- Maintained school servers, backend databases, and legacy Flash-based teaching systems.
- Developed Python-based graphical teaching applications and simple web tools to support interactive learning.

Wuhan Fuya Technology Co., Ltd.

Apr. 2026 - Present

Visual Development Engineer Intern

- Use Comfy UI and built a reusable end-to-end workflow for game character and scene generation using Flux and Canny-based structural control.
- Integrated model experimentation, prompt/control design, and API orchestration into a production-oriented visual generation pipeline.
- Improved the controllability, stylistic consistency, and iteration efficiency of AI-assisted content creation for game art scenarios.
- Contributed to the engineering deployment of generative AI methods in practical visual production workflows.

TECHNICAL SKILLS

- Language: English (fluent, CET-6), Chinese (native)
- Programming and development: Python, Java, C++, JavaScript, SQL, SpringBoot, FastAPI, Redis.
- Machine learning and vision: Deep Learning, Computer Vision, Multimodal Learning, Vision-Language Modeling, Generative Models, Diffusion Models, Controllable Image Generation, RGB-D Visual Modeling, Representation Learning, Feature Fusion, Transformers, Mixture-of-Experts (MoE)
- Research strengths: multimodal feature integration, data preprocessing, model architecture design, backend-system implementation, experimental evaluation.

AWARDS & HONORS

- National First Prize, "Huazhong" Cup National College Mathematical Modeling Competition | **May. 2026**
- National Third Prize, The 7th Global Artificial Intelligence Algorithm Elite Competition Global Finals | **Nov. 2025**
- National Third Prize, National Collegiate Statistical Modeling Contest | **Oct. 2025**
- National Second Prize, The 7th "Huashu" Cup National College Mathematical Modeling Competition | **Aug. 2025**
- National Second Prize in the National Undergraduate Statistical Modelling Competition | **Jun. 2025**
- Provincial Second Prize, iCAN Innovation Contest | **Oct. 2025**
- Provincial Second Prize, The 14th National University Student Market Research and Analysis Competition | **Dec. 2024**
- University Third Prize, The 13th China National Undergraduate "Innovation, Creativity and Entrepreneurship" Challenge | **Dec. 2024**

SOCIAL & SCHOOL ACTIVITIES

University Media & Press Corps

Sep. 2024 – Jun. 2025

Director

- Led campus news production, interviews, and cross-departmental publication coordination.

Big Data Class 12302

Sep. 2024 – Jun. 2025

Academic Representative